José Manuel Requena Plens

Firmware & Software Engineer

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On-site, remote or hybrid

Summary

Embedded firmware and software engineer with experience building industrial firmware for solar inverters (C, STM32, FreeRTOS) and a solid R&D background in acoustics (physics). Focused on low-level optimization, industrial communications (Modbus) and hardware test automation. I maintain *open source* projects used by developers worldwide. Targeting **firmware, software and QA** roles, bringing engineering rigor and a focus on quality.

Experience

R&D Software Developer — **Apr 2023 - May 2026**
Power Electronics S.L. (Solar + Storage) — *Lliria, Spain*

- Reduced RTOS task stack usage by 30% to 50% on STM32 microcontrollers.
- Optimized Modbus TCP request handling, cutting CPU usage under load (40,000 req/s) from 80% to 17% with no dropped responses.
- Redesigned embedded file-system access, cutting access times by over 50%.
- Implemented a scheduled purge (daily/monthly) that safely clears thousands of files, preventing storage exhaustion under continuous operation.
- Built and maintained industrial communications in C with hardware abstraction layers (HAL): Modbus RTU/TCP, UART, I2C and SPI.
- Automated the test suite: 50 to 80 unit tests per module (Ceedling, mocks/fakes, thousands in total) reaching 100% line and branch coverage, plus hundreds of on-hardware integration tests (pytest) for TCP/FTP performance and FTP security, all in CI.
- Reviewed merge requests across a 12-person team, catching bugs before integration.

Predoctoral researcher — **Jul 2021 - Jul 2022**
UPV · CSIC · i3M — *Valencia, Spain*

- Designed acoustic metamaterials (metadiffusers and lenses) that reduce the footprint of conventional diffusers, validated with MATLAB and COMSOL simulations.
- Manufactured and characterized prototypes; results published at Euronoise and Tecniacústica.
- Supervised a Master's thesis in Acoustic Engineering (UPV), from research design to defense.

Researcher in European project — **Feb 2020 - Jul 2021**
UPV · European Space Agency (ESA) · IGIC — *Gandia, Spain*

- Designed and experimentally validated a noise-mitigation metamaterial for the VEGA rocket launch (ESA project AO/1-9479/18/NL/LvH), meeting 100% of project objectives with results accepted by ESA.
- Developed acoustic measurement software per ISO 10534-2 and ASTM 2611 (A|Lab) and a 3-axis robotic system for experimental measurements.
- Co-author of international publications/talks (Euronoise, ECSSMET).

Technical Skills

Embedded Firmware & Electronics: C, STM32 & ARM, ESP32 / ESP-IDF, PlatformIO, Texas Instruments, FreeRTOS / RTOS, Comm. Protocols, HW Debugging, Keil uVision, Instrumentation, Secure Boot & Embedded Security, Doxygen

Software Engineering & Tools: Python, C++, Go, TypeScript, Bash, Clang Tooling, Git, GitHub, GitLab, Docker, MATLAB, CMake, JetBrains Toolbox

Quality, CI/CD & DevSecOps: GitHub Actions, GitLab CI, Jenkins, SonarQube / SonarCloud, GoReleaser, Static Analysis & SAST, Testing, pre-commit hooks

Networking: TCP/IP & Networking, MikroTik RouterOS, WireGuard / VPN, IPv6 & Dual-stack, QUIC / HTTP3, DNS / Dynamic DNS

Security & Cryptography: Applied Cryptography, TLS / mTLS / PKI, Nginx Hardening, CrowdSec, Network Hardening

AI & Agentic Engineering: Model Context Protocol (MCP), AI Coding Agents, LLM Integration

Simulation & Acoustics: COMSOL Multiphysics, LabView, Acoustic Instrumentation

General & Platforms: Linux / Unix, MacOS, Windows, LaTeX, HTML / CSS

Open Source Projects

gitlab-mcp-server (Go · Model Context Protocol server)
51 stars • 59 releases • ~175k downloads • On npm, PyPI & NuGet; listed on Glama, mcp.so, Cursor & mcpservers.org
Open-source GitLab MCP server that exposes 850+ GitLab actions (1,000+ depending on edition and deployment) to AI assistants through a dynamic 2-tool find/execute design, with stdio/HTTP/OAuth transports and safe & read-only modes.

Kleidos (C++ · ESP-IDF · Embedded security)
In development
Password manager firmware for nine M5Stack and LilyGO ESP32 boards: credentials encrypted on the device (AES-256-CBC, HMAC-SHA256, PBKDF2) and typed over Bluetooth LE as a HID keyboard.

phonometry (Python · DSP · Acoustics)
133 stars • 28 releases • PyPI
Standards-conformant acoustics library: sound level metrology, psychoacoustics, room, building and environmental acoustics and vibration, with 1897 CI-enforced conformance checks across 111 domains.

cs-routeros-bouncer (Go · Firewall · MikroTik · WAF)
25 stars • 22 releases • On the CrowdSec Hub
CrowdSec bouncer for MikroTik RouterOS: feeds the collaborative IPS/WAF CrowdSec decisions into the RouterOS firewall (address lists) through its API, blocking malicious IPs in real time.

Education

PhD program in Health and Wellness Technologies (discontinued) — **2019 - 2023**
Polytechnic University of Valencia (i3M / UMIL) — *Valencia, Spain*
Research on acoustic metamaterials for medical imaging and ultrasound beam control.

Master's Degree in Acoustical Engineering — **2019**
Polytechnic University of Valencia (EPSG) — *Gandia, Spain*
Graduated with honors; Master's thesis awarded a special distinction.

Degree in Telecommunication Engineering in Sound and Image — **2018**
University of Alicante (EPS) — *Alicante, Spain*
Two specializations: Acoustic Engineering and Audiovisual Technology.

Superior Sound Technician (Higher Vocational Degree) — **2011**
IES Luis García Berlanga — *Alicante, Spain*

Electronics Technician (Intermediate Vocational Degree) — **2009**
IES Salesianos (San José Artesano) — *Elche, Spain*